

PSN COLLEGE OF ENGINEERING AND TECHNOLOGY, TIRUNELVELI

DEPARTMENT OF AIML

COMPUTER VISION WITH OPENCV

(In association with Eskillite TechServ)

PROJECT ONE PAGE REPORT

Duration:

Date:

**Project Title: VEHICLE-BASED AIR POLLUTION SOURCE
DETECTION**

Name of the team members:

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Problem Statement :

There is a need for an affordable, scalable, and automated system that can monitor vehicular air pollution in real-time using existing camera infrastructure, without deploying costly sensor hardware.

ABSTRACT

Air pollution is one of the most critical environmental challenges of the modern era. Vehicle emissions are a primary contributor to urban air pollution. This project presents a Computer Vision-based approach to detect and classify vehicles from video footage and estimate the level of air pollution caused by the density and type of vehicles present. Using Roboflow for dataset annotation and model training, and Google Colab for inference, the system processes video frames in real-time, identifies vehicle types, assigns pollution weights, and displays a pollution severity level along with a percentage score on each frame of the output video.

The system classifies pollution into five levels: Clean, Low, Medium, High, and Severe based on the cumulative pollution score calculated from detected vehicles. This approach provides a cost-effective, scalable, and automated method to monitor air pollution in urban environments without requiring physical sensors.

Objectives

- To detect and classify vehicles from video footage using a trained deep learning model.
- To assign pollution weights to each vehicle type based on their emission levels.
- To calculate and display a real-time pollution severity level and percentage on the output video.
- To develop a working prototype using Roboflow and Google Colab.

Scope and Purpose of the Project

The main purpose of this project is to develop an automated system for detecting vehicle pollution using object detection techniques. The project aims to identify vehicles emitting excessive smoke from images or videos, thereby helping authorities monitor pollution levels and take necessary actions. This system can be implemented in traffic surveillance systems for real-time monitoring.

The project involves:

- collecting and annotating vehicle pollution datasets using **Roboflow**.
- Training a **YOLO object detection model** in **Google Colab**.
- Performing real-time helmet detection on images and videos.
- Deploying the trained model through an interactive **Gradio web application** for easy access and demonstration.

System Architecture

The system consists of the following components:

- Dataset Collection and Annotation using Roboflow Universe
- Model Training using Roboflow 3.0 (YOLOv8-compatible)
- Video Inference using Google Colab and Roboflow API
- Pollution Scoring and Level Classification
- Output Video Generation with Labels

Expected Outcome

The expected outcome of this project is an automated vehicle pollution detection system capable of identifying smoke-emitting vehicles from images or videos with good accuracy. The system will help in continuous monitoring of air pollution and support environmental protection initiatives.

Dataset

The dataset used is the 'Air Pollution Vehicle' dataset from Roboflow Universe, containing over 3,000 annotated images of vehicles in real traffic scenarios. The dataset includes five vehicle classes:

Vehicle Class	Pollution Weight
Two Wheelers	1
Three Wheelers	2
Passenger Cars	2
Light Commercial Vehicle	3
Multi Axle Goods Vehicles / Buses	4

Pollution Level Calculation

For each video frame, a pollution score is calculated as:

Pollution Score = Sum of (Vehicle Weight x Count for each class)

The pollution percentage is calculated as:

$$\text{Pollution \%} = \min((\text{Score} / 50) \times 100, 100)$$

Score Range	Level	Color
0	Clean	Green
1 - 5	Low	Yellow
6 - 10	Medium	Orange
11 - 15	High	Red
16+	Severe	Dark Red

Future Enhancements

- Real-time implementation using CCTV cameras.
- Integration with IoT devices for smart city applications.
- Automatic alert generation for highly polluting vehicles.
- Deployment as a web or mobile application for easy accessibility.

Tools and Technologies

Tool / Technology	Purpose
Roboflow	Dataset annotation and model training
Google Colab	Cloud-based inference and video processing
Python	Programming language
OpenCV (cv2)	Video frame processing and annotation
Roboflow API	Model inference on video frames
FFmpeg	Output video encoding (H.264)

Implementation Steps

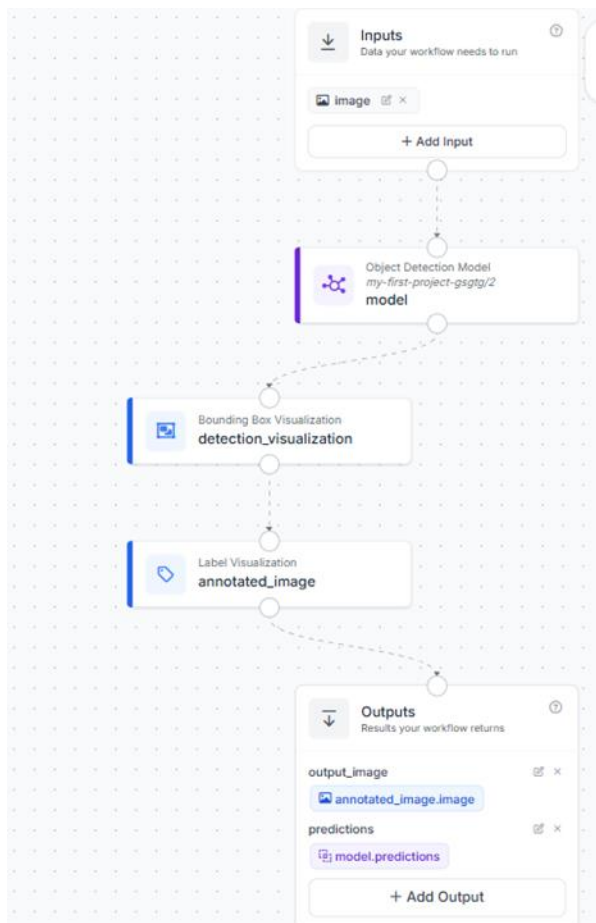
- Collected and annotated the vehicle dataset using Roboflow Universe.

- Trained the model on Roboflow with 3,005 images using Roboflow 3.0 architecture.
- Set up Google Colab environment and installed required libraries.
- Loaded the trained model using the Roboflow API.
- Processed the input video frame by frame using OpenCV.
- Applied model inference on each frame to detect vehicles.
- Calculated pollution score and level per frame.
- Annotated each frame with bounding boxes, labels, and pollution level.
- Compiled annotated frames into output video and converted using FFmpeg.

Sample Code

```
pollution_percent = min((pollution_score / 50) * 100, 100)
cv2.putText(frame, f"Pollution: {level}", (30,50), cv2.FONT_HERSHEY_SIMPLEX,
1.2, color, 3)
```

Work Flow :



RESULTS AND DISCUSSION

← Air pollution Vehicle 2

Test

Download Weights

Deploy Model

Model URL:
air-pollution-vehicle-55jqd/2

Updated On:
Jun 25, 2026, 8:22 PM

Checkpoint:
-

Model Type:
Roboflow 3.0 Object Detection (Fast)

Dataset Version:
2026-06-25 7:50am

Metrics

Valid Set

External

mAP@50
65.5%

Precision
73.5%

Recall
59.1%

F1
65.5%

Test Set

View Test Set

Drop Files or

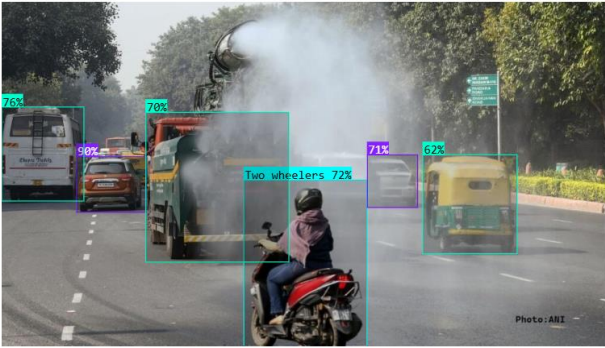
+ Select Files

Image or Youtube URL...

Try with:

My Webcam

My Machine



6 objects detected

Model Visualizations

Confidence Threshold: 50%

Overlap Threshold: 50%

Opacity Threshold: 75%

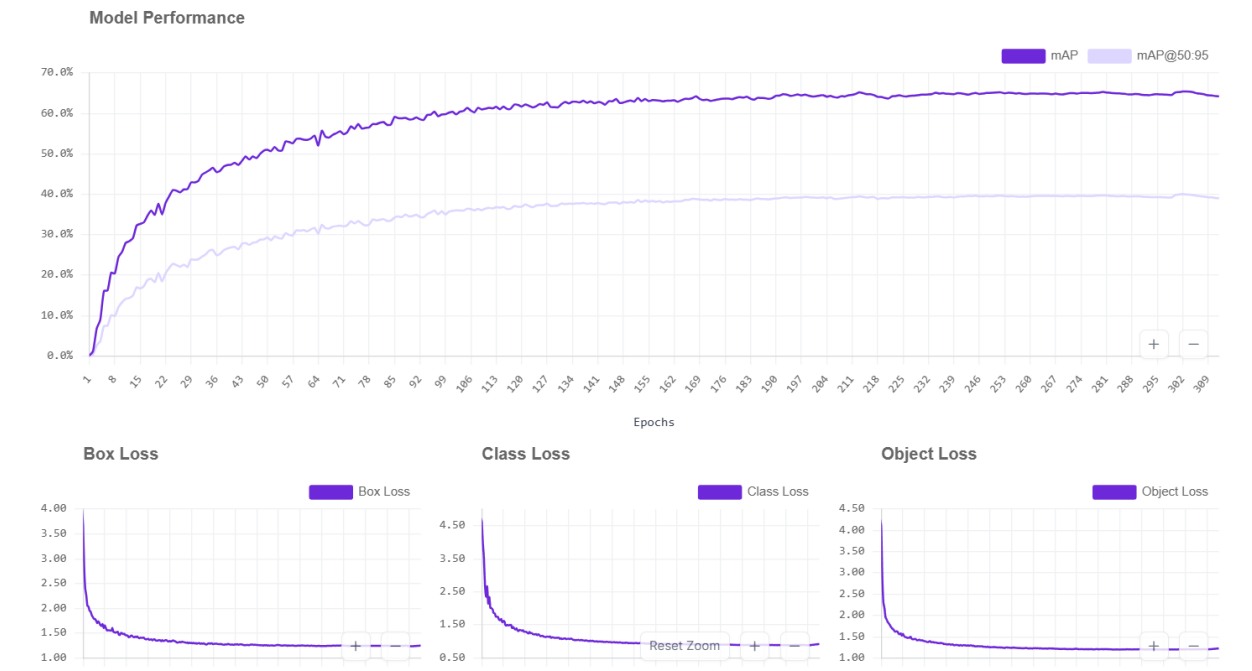
Label Display: Draw Confidence

Show Ground Truth

Output

```
{
  "predictions": [
    {
      "x": 185.5,
      "y": 305.5,
      "width": 117,
      "height": 93,
      "confidence": 0.903,
      "class": "Passenger",
      "class id": 2,
    }
  ]
}
```

Training Graphs

[Advanced Graphs](#)

Output

The system successfully processed the input video and produced an annotated output video. Each frame of the output video displays bounding boxes around detected vehicles with class labels. The pollution level and percentage are displayed prominently at the top of each frame. In the test video containing heavy traffic on a foggy road, the system detected multiple passenger cars and assigned a Severe pollution level with a score of 44, corresponding to approximately 88% pollution level.

Performance

Metric	Value
Dataset Size	3,005 images
Training Split	70% Train / 20% Valid / 10% Test
Model Architecture	Roboflow 3.0 (YOLOv8-compatible)
Detection Confidence Threshold	40%
Pollution Levels Supported	5 (Clean, Low, Medium, High, Severe)
Output Format	MP4 Video (H.264)

Conclusion

This project successfully implemented a Vehicle-based Air Pollution Source Detection system using Computer Vision. Using Roboflow for model training and Google Colab for inference, the system detects vehicles from video footage, classifies them by type, and estimates air pollution levels in real-time. The output video provides a clear visual representation of pollution severity, making it a practical tool for urban air quality monitoring.

Inference

The project demonstrates that object detection techniques can be effectively used to identify vehicle pollution sources. The results indicate that automated monitoring systems can improve pollution management and support the development of cleaner and smarter cities.

Signature of the student

Signature of the HoD

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